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Editor-in-Chief
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Dear Editor-in-Chief,

I am pleased to submit the manuscript “**LLM-Based Test Case Generation from Natural-Language Requirements: A Prompt-Engineered, Rule-Verified Framework with Executable Mutation Validation**” for consideration as a Regular Research Article in the *Journal of Systems and Software*.

Contribution. The paper presents RAITG (Requirement-Aware Intelligent Test Generator), an LLM pipeline that converts natural-language requirements into executable test cases under a prompt-engineered, rule-verified framework. The pipeline is evaluated against *executable* mutation testing rather than self-reported quality scores, yielding a measured 82.99% mutation kill rate across 312 requirements and 936 generation runs spanning three industrial domains (commercial web, financial services, healthcare). The manuscript includes a full prompt-engineering ablation, rule-verification failure mode taxonomy, cross-domain transferability analysis, executable test-suite statistics, and a deployment cost model.

Fit for JSS. The contribution is a complete systems-and-software research article: it presents the framework architecture, the executable validation methodology, the empirical evidence across three industrial domains, threats-to-validity analysis, and a path from research prototype to deployable tool. JSS readers concerned with practical software-engineering tooling, LLM-based development support, and rigorous evaluation methodology are the intended audience.

Reproducibility. The RAITG dataset (312 requirements and 936 generation runs across three domains) is permanently archived on Zenodo with DOI [10.5281/zenodo.20285104](https://doi.org/10.5281/zenodo.20285104). Source code, prompts, rule-verification scripts, and the mutation-testing harness are public at https://github.com/javvadivijayprasad/TestCaseGen_Reserach.

Honest disclosure. The reference applications used for executable validation are research-scale rather than production-scale, and the LLM under evaluation is a single commercial provider (Anthropic Claude). Both limitations are stated in the Threats to Validity section. Cross-provider generalisation and large-application validation are explicitly framed as future work.

Originality and exclusivity. This manuscript has not been published previously and is not under consideration at any other journal or conference.

Use of AI tools. In preparation of this manuscript, I used Anthropic’s Claude (a large language model) for drafting assistance and copy-editing of the prose. All empirical claims, data, model

training, and statistical analyses were generated by my own scripts and verified against the on-disk artifacts; the AI tool was not used to generate data or results. The same LLM is also the system under evaluation in the experimental work; this dual role is fully disclosed in the manuscript's Acknowledgments section and discussed under Threats to Validity.

Conflicts of interest. None.

Funding. No external funding was received.

Suggested associate editor topics. Software engineering, LLMs for software engineering, test case generation, empirical evaluation, mutation testing.

I would be grateful for your editorial consideration and look forward to the reviewers' feedback. Thank you for your time.

Sincerely,

Vijay Prasad Javvadi